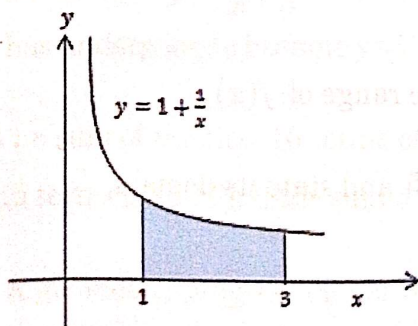


Exam Practice Papers

Paper 1A

- 1) a) Express $-3x^2 - 12x + 21$ in the form $a + b(x + c)^2$ where a, b and c are constants to be found. [3]
- b) Sketch the graph of $y = -3x^2 - 12x + 21$ showing the coordinates of the turning point. [2]
- 2) Find the x coordinates of the turning points of the curve $y = x^3 - 2x^2 - 5x + 6$ in exact form and determine their nature. [6]
- 3) a) Given that $y = \frac{x^2 + x}{\sqrt{x}}$ find
- i) $\frac{dy}{dx}$ [2]
- ii) the equation of the tangent to the curve at $x = 1$. [2]
- b) Differentiate $(2x^2 + 3)^4$ with respect to x . [2]
- 4) a) Show that $\frac{\sin A + \sin 2A}{1 + \cos A + \cos 2A} = \tan A$. [3]
- b) Hence solve the equation $\frac{\sin A + \sin 2A}{1 + \cos A + \cos 2A} = -\frac{3}{4}$ for $-180^\circ \leq A \leq 180^\circ$ [3]
- 5) Given that $x+2$ is a factor of $f(x) = 2x^3 + ax^2 - 13x + b$ and that -18 is the remainder when $f(x)$ is divided by $x - 1$,
- a) find the values of a and b . [4]
- b) factorise $f(x)$ completely. [3]
- 6) a) Express $\frac{x^2}{4 - x^2}$ in partial fractions. [3]
- b) The diagram below shows the graph of $y = 1 + \frac{1}{x}$ for positive values of x .



Find the volume generated when the shaded area is rotated 360° about the x -axis. [4]

7) The position vectors of A, B, C and D are respectively $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$.

Find

a) $\overrightarrow{AB}$ [1]

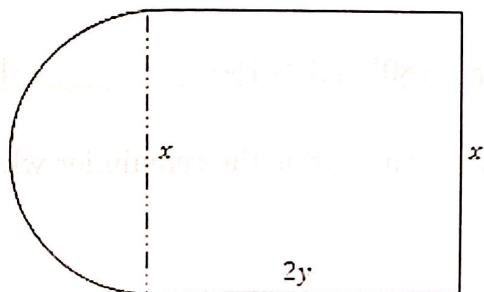
b) the acute angle between the directions of $\overrightarrow{OA}$ and $\overrightarrow{OB}$. [4]

c) $\overrightarrow{OB} \cdot \overrightarrow{OD}$ and state the angular relationship between the two vectors. [3]

d) the unit vector in the direction of BC. [2]

8) a) Rationalise the denominator of $\frac{3 + \sqrt{5}}{2 + \sqrt{3}} - 1$. [3]

b) An enclosure is made up of a rectangle measuring x metres by $2y$ metres. At one end of the rectangle, and using one side of the rectangle as a diameter (see diagram), a semi-circle is joined to the rectangle. The perimeter of the enclosure is to be fenced using a total of 90m of diamond mesh wire.



i) Express y in terms of x . [2]

ii) Find the value of x for which the area enclosed is maximum. [5]

9) a) You are given the functions $f(x) = \sin x$: $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$; $g(x) = \frac{1}{x}$: $x \neq 0$ and $h(x) = 2x - 1$

Find i) $f \circ h$ ii) $h \circ f \circ g$ iii) $h \circ h \circ h$ iv) the range of $f(x)$. [8]

b) Find the inverse of the function $p(x) = 2x^2 - 4x - 5$ and state its domain. [3]

10) a) If $z = 5 - i$ and $w = 1 + 2i$,

i) Find $\frac{z}{w}$, $\arg\left(\frac{z}{w}\right)$ and $\left|\frac{z}{w}\right|$ [4]

ii) Represent w and $\bar{w}$ on an Argand diagram. [2]

b) Find x and y if $x + y + i(x - y) = 5 + 3i$ [3]

c) Simplify $(1 + \sqrt{3}i)^5 - (1 - \sqrt{3}i)^5$. [3]

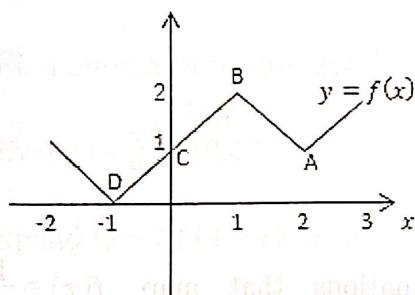
11) a) A curve has equation $2y^3 + y^2 - y^5 = x^4 - 2x^3 + x^2$.

i) At how many points does this curve have horizontal tangent lines? [2]

ii) Find the x coordinates of the points where the curve has horizontal tangent lines. [5]

b) Solve the inequality $\frac{x}{(x-3)(x+1)} \leq -\frac{2}{5}$. [5]

12) a)



The diagram above shows the graph of $y = f(x)$.

Sketch the graph of $y = f(x)$ after the given transformations. In each case show, or state, the images of A, B, C and D.

i) $y = f(x - 2)$ ii) $y = -2f(x)$ iii) $y = f(x) + 1$

iv) $y = f\left(\frac{1}{2}x\right)$ v) $y = -f(x - 2)$ [10]

b) Given that $y = \frac{1-x}{x+3}$, list, in correct order the transformations that the graph of $y = \frac{1}{x}$ has undergone to become $y = (1-x)/(x+3)$. [4]

13) a) The sum of the first 10 terms of an arithmetic progression is 120 and the sum of the first 20 terms is 840 Find the sum of the first 40 terms of this progression. [5]

b) A geometric progression has 8th term $\frac{2}{3}$ and 5th term 18.

Find the sum to infinity of the geometric progression.

[5]

- c) Use a geometric progression to express the recurrent decimal number 0.45252 as a fraction in its lowest terms.

[4]